**24CS2506 – Building Applications using GPT-4**

**Experiment 4: Personalized Recommendation System**

## 1. What is a Recommendation System?

A **Recommendation System** is an AI-based system that suggests items, content, products, services, or information to users based on their preferences, interests, previous interactions, or requirements.

**Examples**

- Netflix → Movie recommendations
- YouTube → Video recommendations
- Amazon → Product recommendations
- Spotify → Music recommendations
- Online learning platforms → Course recommendations

**Simple Example**

**User Preference:**

Interested in Python, Artificial Intelligence and Data Science.

**Recommendation:**

1. Python for Data Science
2. Machine Learning Fundamentals
3. Generative AI with Python
4. Data Analytics using Python
5. Deep Learning Fundamentals

## 2. What is Personalized Recommendation?

A **personalized recommendation** provides different suggestions for different users.

For example:

**Student A**

Interests:

Python, AI, Data Science

Recommendations:

Python Programming
Machine Learning
Data Science Projects

**Student B**

Interests:

Cybersecurity, Networks, Ethical Hacking

Recommendations:

Network Security
Ethical Hacking
Cybersecurity Fundamentals

Therefore:

**User Profile → Preferences → AI Analysis → Personalized Recommendations**

## 3. Why Use Generative AI for Recommendation?

Traditional recommendation systems generally depend on predefined rules, ratings, or similarity calculations. Generative AI can understand **natural-language descriptions of user preferences**.

For example:

"I am a beginner who likes programming and wants to learn AI. I prefer practical courses rather than theoretical courses."

An LLM can understand this description and generate suitable recommendations.

**Advantages**

- Understands natural language
- Can analyze user preferences
- Generates explanations for recommendations
- Can consider multiple interests simultaneously
- Can personalize recommendations
- Can interact conversationally with users

**4. What is GPT-4?**

GPT-4 is a Generative AI language model capable of understanding and generating natural language.

In a recommendation application, GPT-4 can be used to:

- Analyze user preferences
- Identify interests
- Understand user requirements
- Generate recommendations
- Explain why an item was recommended
- Refine recommendations based on feedback

**5. Understanding User Profile**

A recommendation system requires information about the user.

**Example User Profile**

| Parameter | Example |
|---|---|
| Name | Arun |
| Age | 20 |
| Field | Computer Science |
| Interests | AI, Python |
| Skill Level | Beginner |
| Preferred Type | Practical |
| Goal | Learn AI |

The system can convert these inputs into a structured prompt.

**6. Feedback-Based Recommendation**

The system can improve recommendations based on feedback.

**Initial Recommendation**

Python Programming
Machine Learning
Web Development

**User Feedback**

"I am more interested in AI than Web Development."

**Updated Recommendation**

Artificial Intelligence
Machine Learning
Deep Learning
Generative AI
Computer Vision

This demonstrates **personalization through interaction**.

**7. Example Expected Output**

**Personalized Recommendations**

**User:** Computer Science student
**Interests:** AI, Python
**Level:** Beginner

| Rank | Recommendation | Reason | Score |
|---|---|---|---|
| 1 | Python for AI | Matches Python and AI interests | 95% |
| 2 | Machine Learning Basics | Suitable for beginner AI learners | 92% |
| 3 | Generative AI | Relevant to current AI interests | 88% |
| 4 | Data Science with Python | Combines Python and data analysis | 85% |
| 5 | Deep Learning Basics | Natural progression after ML | 80% |